B-cell epitope prediction from antigen-only dynamics: project log

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Living record. Read top to bottom to get up to speed.

Running record of the project: predict B-cell epitope residues from *antigen-only* information, using alanine-scan $\Delta\Delta G_{\text{bind}}$ as residue-level binding-importance labels and asking whether antigen dynamics / solvation / energetics carry predictive signal *beyond static surface exposure*. Read in order; each section is one iteration — what we tried, what we found, what it changed. Not a formal manuscript. Every figure is regenerated by a named script (recorded in `figures/FIGURES.md`); every number traces to a file or command; unconfirmed values are flagged `[verify]`.

1 Motivation and the gap

We want to predict which residues of an antigen are B-cell epitope residues, from the antigen structure alone, in cases where the antibody is unknown. Doing this rigorously requires residue-level labels for "binding importance." We take these from **alanine scanning**: for residue i, $\Delta\Delta G^{(i)}_{\text{bind}} = \Delta G^{\text{mut},i}_{\text{bind}} - \Delta G^{\text{WT}}_{\text{bind}}$, with large positive values flagging residues whose wild-type identity stabilizes the interface. We treat this as a residue-level epitope score, $S^{\text{epitope}}_i \approx \Delta\Delta G^{(i)}_{\text{bind}}$. (Full derivation and the alchemical / StaB-ddG label tiers: `../../tcr/docs/b_cell_eptiope_prediction_may23_update_final.pdf`

The gap. Most B-cell epitope predictors — and the benchmarks that train them — label residues by *contact distance* to a bound antibody. As a coarse first pass this is defensible: proximity to the paratope is a reasonable way to *nominate* candidate interface residues. The limitation is treating every contact as a positive label of equal weight, which conflates "close" with "important": many interface residues contribute little binding free energy, while a few hotspots carry most of the affinity. The bias is baked into the data itself. Of the 4,164 conformational (discontinuous) B-cell epitopes in IEDB [1] that carry a deposited structure, essentially all (4,161; $> 99\%$) are defined by crystallographic or cryo-EM *contact* rather than by any binding measurement;[1] models trained on these labels therefore inherit proximity as their working definition of "epitope." Alanine-scan $\Delta\Delta G$ supplies the missing energetic refinement — *which* of the contacting residues actually pay for binding. Even a binary critical/non-critical call from a saturating alanine scan is energetically grounded, since "critical" means mutation measurably weakens binding; proximity and energetics are thus complementary, the former a candidate filter and the latter the importance label.

Separately, since Westhof et al. [2] it has been known that antigenicity correlates with *segmental mobility* (flexibility / B-factor) — an antigen-intrinsic, antibody-agnostic property — and

[1] IEDB next-generation query API (`query-api.iedb.org`), `bcell_search`, structure type *Discontinuous peptide* with a non-null `pdb_id`, queried 2026-06. IEDB attaches a structure only when the epitope was itself defined from one, so structure-linked records are overwhelmingly contact-derived; the mutation/binding-derived epitopes carry no `pdb_id`.

energy-decomposition analyses of plain MD trajectories can already localize antibody epitopes on an antigen without knowledge of the antibody [3]. The opportunity is to test, with modern MD, whether antigen-only *dynamical* and *solvation* features (RMSF, collective-mode participation, hydration thermodynamics) predict energetic binding importance — and whether they beat static exposure. This is where the author's methods background (FRESEAN collective-mode analysis; 3D-2PT-style hydration thermodynamics; non-equilibrium energy flow) is the differentiator (`../../_literature/my_manuscripts/`).

2 Design decisions to hold onto

Five issues decide whether the project produces a result or an artifact. Recorded here so they stay front-of-mind.

1. **The label is antibody-specific; the features are antibody-agnostic.** $\Delta\Delta G^{(i)}_{\text{bind}}$ is a residue's contribution to *one* paratope, but $\mathbf{x}_i$ comes from the apo antigen and cannot know the antibody. So $f(\mathbf{x}_i) \to \Delta\Delta G^{(i)}$ can only recover the *antibody-marginal* (antigen-intrinsic) hotspot propensity. This is not a weakness to hide — it is exactly why dynamics/flexibility features should help (cf. Westhof). Where an antigen has several antibodies, average $\Delta\Delta G^{(i)}$ over them to *define* the intrinsic target.

2. **"Beyond SASA" is the primary hypothesis, not a footnote.** Surface exposure is the master confounder: exposed residues are simultaneously more likely to be hotspots, more flexible, and more hydrated. The publishable claim is not "flexibility predicts epitopes" (Westhof has that) but "dynamics/solvation features add signal *after* controlling for static exposure." Design: baseline = SASA + depth + residue type; then test the partial R^2 added by RMSF, mode participation, and 3D-2PT hydration.

3. **Effective $N \approx$ #antigens/folds, not #residues.** Residues within an antigen are correlated and the intrinsic signal is per-fold. Split **leave-one-complex-out** (ideally leave-one-fold-out); never random residue splits. At this scale the work is hypothesis-generating — powered for a strong dynamics-vs-exposure effect, not a deployable predictor. State this plainly.

4. **Features come from the apo antigen** — which is the point and sets a ceiling. Cryptic / induced-fit epitopes (conformational selection vs. induced fit) cannot be predicted in principle from apo features. Conformational-selection epitopes *should* show in apo flexibility/mode features; pure induced-fit ones will not. That contrast is itself a result.

5. **The MD protocol is not one stack.** A flexibility/RMSF run differs from a FRESEAN/3D-2PT run (the latter needs high-frequency velocity output, constraint removal, and a water model validated for dynamics). Likely two protocols per antigen. The house TIP3P/2 fs/LINCS default is scrutinized accordingly (see `../docs/METHODS.md`).

3 The label benchmark

Curated experimental alanine-scan $\Delta\Delta G$ for antigen-side residues lives in `benchmark/antigen_alanine_scanning` (6 sheets: README, Simple_Primary, Full_Schema [51 cols], hGH_Table2_4, Paper_Inventory, Counts). Audited 2026-06-18.

Coverage. 203 primary rows, 13 datasets, 7 antigens. Source: AB-Bind [4] (195 rows; `github.com/sarahsirin/A` and the Jin hGH alanine scan [5] (8 rows; benchmark cites the 1994 thesis, Table 3.2). The dataset was subsequently expanded with three antigen-side scans from SKEMPI 2.0 [6] and a binary label tier from IEDB (§5 and the Tier-2 section). Sign convention: $+\Delta\Delta G \Rightarrow$ weaker binding $\Rightarrow$ residue important (38 negative = binding-improving alanines; 17 zeros).

Integrity. The $\Delta\Delta G \leftrightarrow K_d$-ratio arithmetic is internally consistent for all 203 rows ($< 2\%$ vs $\exp(\Delta\Delta G/RT)$). No censoring in the primary set. The 4 non-alanine rows (hGH, EC50-derived) are correctly flagged `exclude_from_binding_label`. Provenance (DOI, source line, URL) is complete except the two homology-modeled BoNT/A2 sets and the Jin rows. `transcription_confidence` is uniformly `high` — treat as a soft flag, possibly a default fill. `[verify]` spot-check 1–2 known hotspots against the source papers.

Clean MD-correlation core. Dropping homology models (`HM_`) and EC50/excluded rows leaves **163 rows / 5 antigens on real PDBs** (VEGF, lysozyme, HPr, BoNT/A1, MT-SP1).

4 First system: hen egg-white lysozyme (HEL)

HEL is the cleanest entry point: small (129 residues), rigid (apo ≈ bound), very stable, textbook AMBER behavior, and — crucially — it carries **three** independent antibody alanine scans in the benchmark, on a consistent 1–129 numbering:

Antibody	PDB	Ag chain	# Ala	$\Delta\Delta G$ range (kcal/mol)
HyHEL-63	1DQJ	C	11	0.60–6.10
D1.3	1VFB	C	12	0.20–2.90
HyHEL-10	3HFM	Y	4	1.03–6.83

Why this is better than one antibody. HyHEL-10 and HyHEL-63 share an epitope and *agree* on hotspots (K96: 6.83 vs 6.10; Y20: 4.87 vs 3.30; K97: 6.18 vs 3.50; R21: 1.03 vs 1.30) — direct evidence for the antigen-intrinsic hotspot idea (Design note 1). D1.3 binds a *non-overlapping* epitope (residues 18–24 and 116–129; zero residue overlap with the HyHELs), giving a second epitope plus built-in negative controls. Because we extract features from the apo antigen, **one apo-HEL trajectory** serves all three antibodies and their mean.

Starting structure. RCSB **1AKI** (1.5 Å X-ray, chain A, apo). Verified: the 1AKI sequence matches the benchmark WT identity at all 23 labeled positions, so labels map 1:1 onto 1–129 numbering (K96/K97/Y20 hotspots confirmed). Apo crystal chosen over antigen-stripped-from-complex to match the prediction scenario; HEL's rigidity makes the choice low-risk.

Figure 1: Apo HEL starting structure (RCSB **1AKI**, 1.5 Å X-ray, chain A); cartoon coloured by secondary structure (helices purple, sheets yellow). This single apo trajectory serves all three antibody scans.

5 Simulated antigens

One section per antigen we simulate: what it is, the structural handle we use, and its alanine-scan labels. Each structure is coloured per residue by $\Delta\Delta G_{\mathrm{bind}}$ on a diverging scale with **white pinned at** 0: red = positive (hotspot / likely epitope), blue = negative, grey = residue not scanned. One panel per antibody; a single global colour scale ($\pm\max|\Delta\Delta G_{\mathrm{bind}}|$ over the whole benchmark, white at 0) is shared by every figure, so colours are directly comparable across antigens. Vertical colourbar at right. $\Delta\Delta G_{\mathrm{bind}}$ is the change in binding free energy, K_d/K_i-derived; positive weakens binding.

5.1 Lysozyme (hen egg-white, HEL)

Hen egg-white lysozyme is a 129-residue glycoside hydrolase and the canonical model antigen of structural immunology: multiple high-resolution antibody complexes and alanine scans exist (HyHEL-10, HyHEL-63, D1.3). It is compact and conformationally rigid, cross-linked by four disulfides. We simulate the apo enzyme from `1AKI`. The labels here are the HyHEL-63 scan (`1DQJ`, Table 1) [7]; the v2 benchmark adds D1.3 (`1VFB`) [8] and HyHEL-10 (`3HFM`) [9] scans for the same antigen. These $\Delta\Delta G_{\mathrm{bind}}$ values are drawn from the AB-Bind database [4].

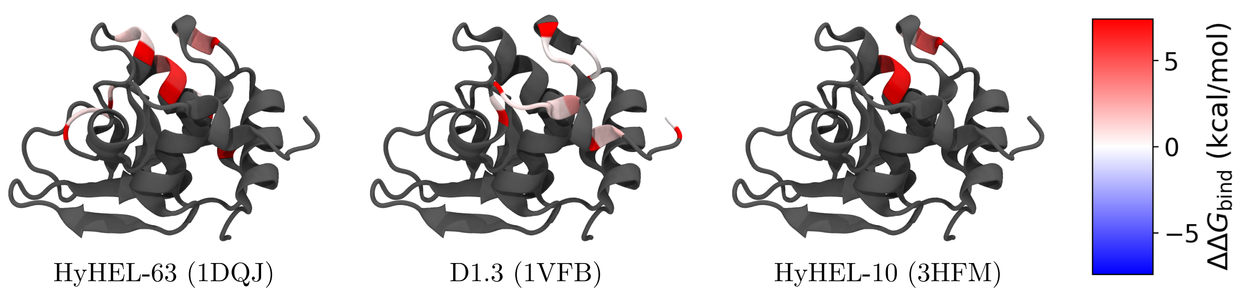


Figure 2: HEL coloured per residue by alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ for its three antibodies (shared scale, white = 0, red = hotspot, blue < 0; grey = not scanned). Distinct but overlapping epitope cores across HyHEL-63, D1.3 and HyHEL-10.

Table 1: **Lysozyme** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: apo 1AKI). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	HyHEL-63 (1DQJ)	IgG1-kappa D1.3 Fv (1VFB)	HyHEL-10 Fv (3HFM)
D18	–	+0.30	–
N19	–	+0.30	–
Y20	+3.30	–	+4.87
R21	+1.30	–	+1.03
Y23	–	+0.40	–
S24	–	+0.80	–
W62	+0.80	–	–
W63	+1.30	–	–
L75	+1.50	–	–
T89	+0.80	–	–
N93	+0.60	–	–
K96	+6.10	–	+6.83
K97	+3.50	–	+6.18
S100	+0.80	–	–
D101	+1.30	–	–
K116	–	+0.70	–
T118	–	+0.80	–
D119	–	+1.00	–
V120	–	+0.90	–
Q121	–	+2.90	–
I124	–	+1.20	–
R125	–	+1.80	–
L129	–	+0.20	–

5.2 HPr

The histidine-containing phosphocarrier protein (HPr) is a small (∼9 kDa, 85-residue) cytoplasmic protein of the bacterial phosphoenolpyruvate:sugar phosphotransferase system, shuttling a phosphoryl group between enzyme I and the sugar-specific EIIA components. The *E. coli* protein is a well-folded open-faced β-sandwich with *no cysteines*, making it an exceptionally clean dynamics test case. The Jel42 epitope clusters in the C-terminal half (Table 2) [10]; $\Delta\Delta G_{\text{bind}}$ values via AB-Bind [4]. We simulate the apo protein from `2JEL` (chain P).

Figure 3: HPr coloured per residue by Jel42 alanine-scan $\Delta\Delta G_{\text{bind}}$ (white = 0, red = hotspot; grey = not scanned). A single dominant hotspot (E70).

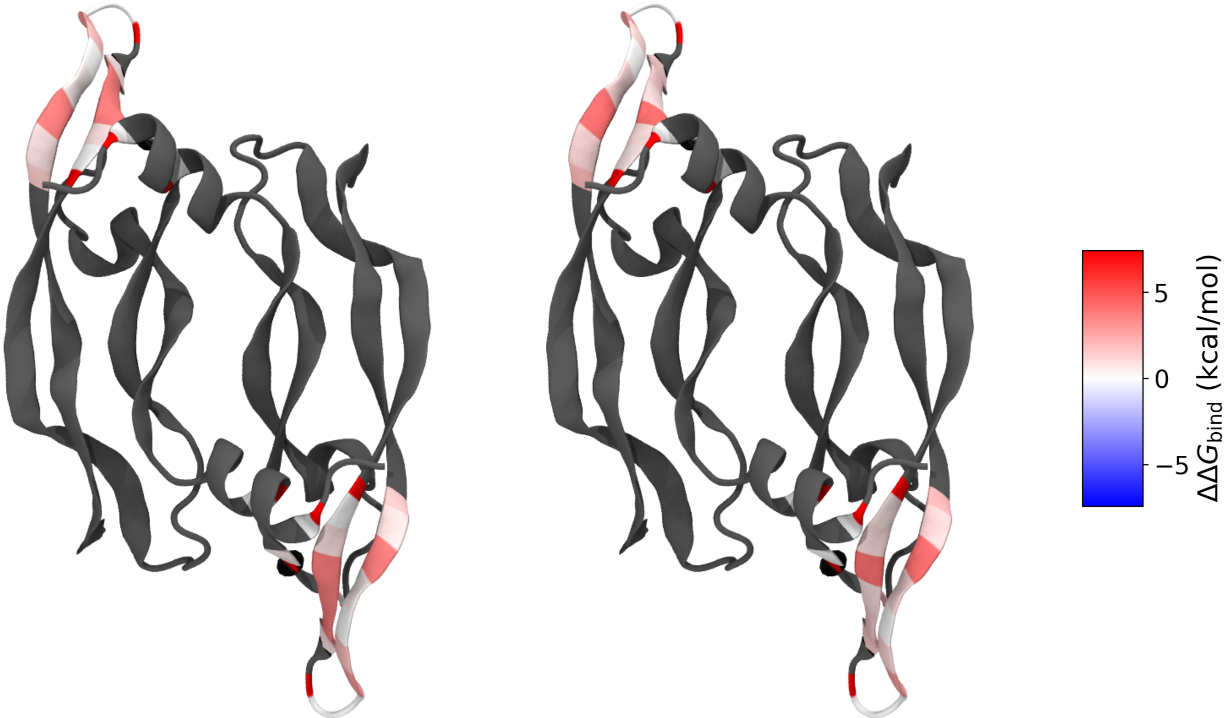
Table 2: **HPr** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; competition, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 2JEL chain P). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	Jel42 (2JEL)
T62	+0.00
E68	+0.41
E70	+2.75
H76	-0.41
E83	+0.00
E85	+0.00

5.3 VEGF-A

Vascular endothelial growth factor A is a secreted pro-angiogenic cytokine and a central drug target (bevacizumab, ranibizumab). Its receptor-binding domain is an antiparallel, disulfide-linked homodimer with a cystine-knot growth-factor fold; the two protomers are covalently joined, so the biological unit must be simulated as the dimer. The benchmark pairs two antibodies against the same receptor-binding face: fab-12 (`1BJ1`, the bevacizumab precursor) [11] and the affinity-matured

Y0317 (`1CZ8`, ranibizumab) [12] — a useful same-antigen/different-affinity comparison (Table 3); $\Delta\Delta G_{\mathrm{bind}}$ values via AB-Bind [4]. We simulate the apo dimer from `1BJ1` (chains V, W).



fab-12 (1BJ1)

Y0317 (1CZ8)

Figure 4: VEGF-A homodimer coloured per residue by $\Delta\Delta G_{\mathrm{bind}}$ for each antibody (shared scale, white = 0; grey = not scanned). Both protomers carry the same labelled residues; red marks the hotspots.

5.4 MT-SP1 (matriptase)

MT-SP1 / matriptase (ST14) is a type II transmembrane serine protease of the S1 family; its extracellular catalytic domain is over-expressed in epithelial cancers and is a validated oncology target. The benchmark uses two inhibitory Fabs, E2 (`3BN9`) [13] and S4 (`3NPS`) [14], scanning the protease domain in chymotrypsin numbering (Table 4; $\Delta\Delta G_{\mathrm{bind}}$ via AB-Bind [4]). Simulation is staged but **pending**: the crystal structures lack two internal residues (149, 218) that must be repaired before MD (§9).

`figures/mtsp1_structure.png`
(not yet regenerated; see figures/FIGURES.md)

Figure 5: MT-SP1 protease domain (3NPS chain A). *Placeholder.*

Table 3: **VEGF** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1BJ1 chains V,W (dimer)). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

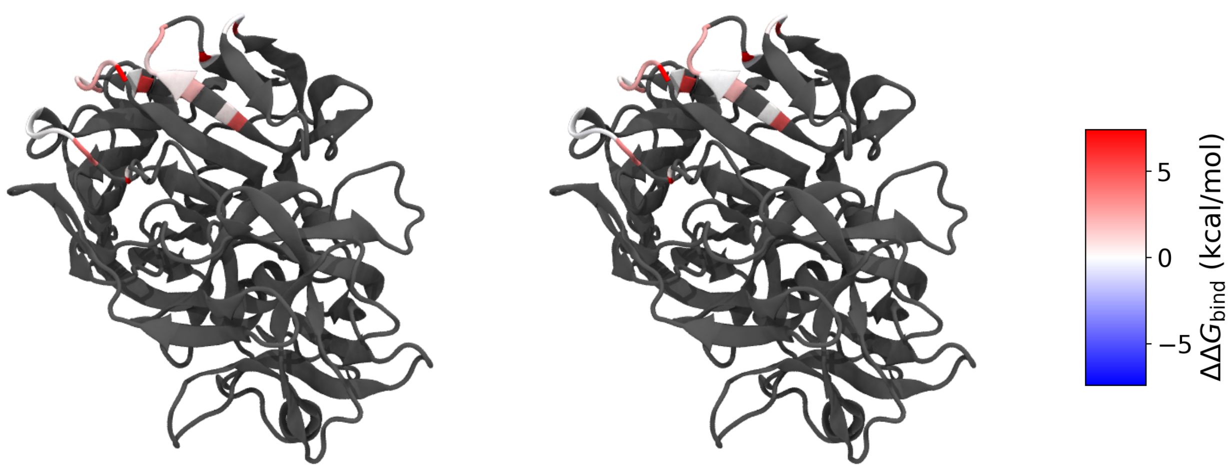
Residue	fab-12 (1BJ1)	Y0317 (1CZ8)
F17	+0.00	+0.00
Y21	+0.00	+0.00
Y45	+0.82	+1.94
K48	+0.41	+0.00
Q79	+0.00	+0.65
I80	+0.82	+0.95
M81	+3.69	+4.10
R82	+3.69	+0.82
I83	+3.69	+1.30
K84	+0.65	+1.36
H86	+0.00	+0.00
Q87	+0.00	+0.00
G88	+2.76	+2.67
Q89	+1.75	+1.06
H90	+0.00	+0.00
I91	+0.41	+1.06
G92	+3.69	+4.10
E93	+0.82	+1.15
M94	+1.42	+1.91

5.5 Bont/A1 (H_C receptor-binding domain)

Botulinum neurotoxin serotype A1 is among the most potent toxins known: a ∼150 kDa di-chain protein comprising a zinc-metalloprotease light chain and a heavy chain. Its C-terminal heavy-chain domain (H_C, here residues 872–1295) is the receptor-binding domain — engaging neuronal gangliosides and the SV2 protein — and the principal protective antigen targeted by neutralizing antibodies. We simulate the isolated H_C domain (`2NYY` chain A); the catalytic light-chain zinc lies outside this fragment. Two neutralizing mAbs, CR1 (`2NYY`) and AR2 (`2NZ9`), provide the labels (Table 5) [15]; $\Delta\Delta G_{\text{bind}}$ via AB-Bind [4].

Table 4: **MT-SP1** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; enzymatic inhibition, K_i-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 3NPS chain A). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	E2 (3BN9)	S4 (3NPS)
Q23	–	+0.03
I26	–	+0.65
Q38	-0.42	–
I41	+0.00	–
I45	–	-0.34
D46	–	+0.34
D47	–	+1.08
R48	–	-1.07
F50	–	+0.22
R51	–	+0.14
Y52	–	+0.46
I60	+0.84	–
R82	–	-0.15
R87	-0.16	–
F89	–	+1.61
N90	–	+0.26
D91	–	+1.52
F92	–	+0.47
T93	–	+0.73
F94	+0.64	–
N95	+0.77	–
D96	+6.70	–
F97	+6.70	–
T98	+1.13	–
H138	–	+1.89
Q140	–	+0.30
Y141	–	+1.79
H143	+0.09	–
T144	–	+0.18
Q145	+0.13	–
Y146	+1.09	–
L147	–	+0.30
T150	+0.29	–
L153	+0.34	–
E163	–	+0.62
Q168	–	-0.06
E169	+0.37	+0.75
Q174	-0.03	–
Q175	+2.51	–
D214	–	+1.48
D217	+0.57	–
Q218	–	-0.04
R219	–	-0.08
K221	–	-0.10
R222	-0.09	–
K224	+0.79	–



CR1 (2NYY) AR2 (2NZ9)

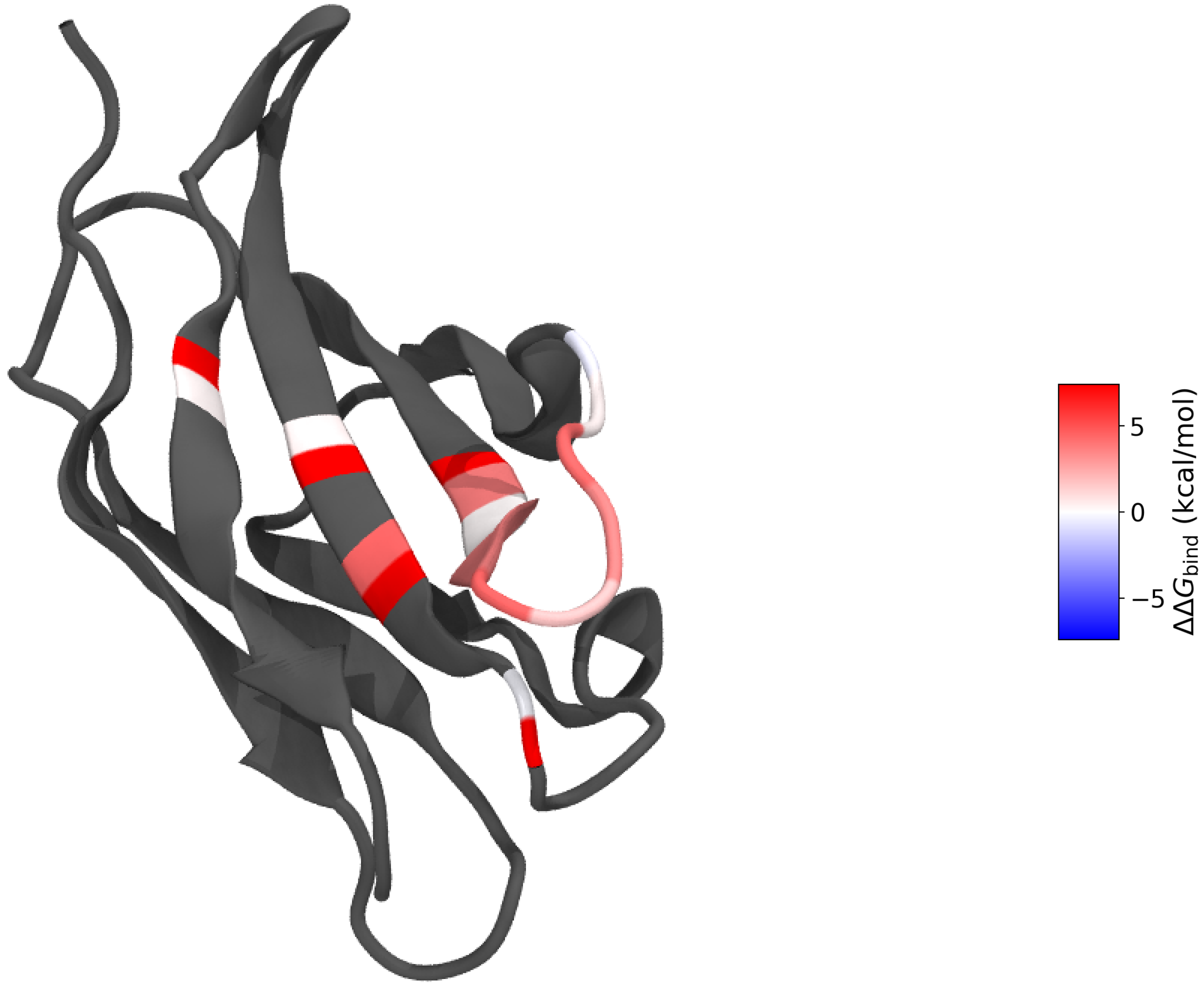
Figure 6: Bont/A1 H_C coloured per residue by $\Delta\Delta G_{\text{bind}}$ for two neutralizing mAbs (shared scale, white = 0; grey = not scanned). Red marks the hotspots — the strongest reach $\Delta\Delta G_{\text{bind}} > 7$ (binding essentially abolished).

Table 5: **Bont/A1** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; KinExA, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: extract 2NYY Hc (res 872-1295)). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	CR1 (2NYY)	AR2 (2NZ9)
S902	-0.23	-0.12
K903	+0.29	+0.54
Q915	+0.52	+0.10
F917	+0.42	-0.05
N918	+0.89	+2.16
L919	+2.59	+2.28
E920	+2.84	+2.77
K923	-0.14	-0.30
F953	+4.06	+3.34
N954	+0.10	-0.15
S955	+0.00	-0.08
I956	-0.01	+0.07
K1056	-0.03	-0.01
D1058	+0.01	-0.03
R1061	+0.82	+0.29
D1062	+2.53	+2.34
T1063	+1.62	+2.37
H1064	+7.32	+7.42
R1294	+0.30	+0.39

5.6 IFN-γ receptor

The interferon-γ receptor α-chain (IFN-γR1) ectodomain is the high-affinity ligand-binding subunit of the type II interferon receptor: two fibronectin type III domains forming an L-shaped β-sandwich that engages the IFN-γ homodimer. The neutralizing antibody A6 binds this ectodomain and the published alanine scan localises a compact energetic hotspot on one face (Table 6) [16]; $\Delta\Delta G_{\mathrm{bind}}$ values curated via SKEMPI 2.0 [6]. It is a clean, rigid, single-chain ~95-residue Ig-like domain — an attractive small test antigen. We simulate the apo ectodomain stripped from `1JRH` (chain I).



A6 (1JRH)

Figure 7: IFN-γ receptor coloured per residue by A6 alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned). The hotspots cluster on a single β-sandwich face.

5.7 Tissue factor

Tissue factor (TF, coagulation factor III) is the cell-surface initiator of the extrinsic blood-coagulation cascade: its extracellular region is a two-domain fibronectin type III module that binds and allosterically activates factor VIIa. The inhibitory Fab 5G9 blocks the macromolecular substrate site, and its antigen-side alanine scan marks the functional epitope (Table 7) [17]; $\Delta\Delta G_{\mathrm{bind}}$ values curated via SKEMPI 2.0 [6]. The ectodomain is a well-folded, single-chain ~200-residue β-rich protein. We simulate the apo ectodomain stripped from `1AHW` (chain C).

Table 6: **IFN-gamma receptor** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1JRH chain I). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	A6 (1JRH)
K47	+3.71
N48	+0.17
Y49	+3.53
G50	+4.45
V51	+1.68
K52	+3.39
N53	+4.30
S54	+0.38
E55	-0.57
N79	-0.20
W82	+4.43
R84	+0.39
K98	+0.31

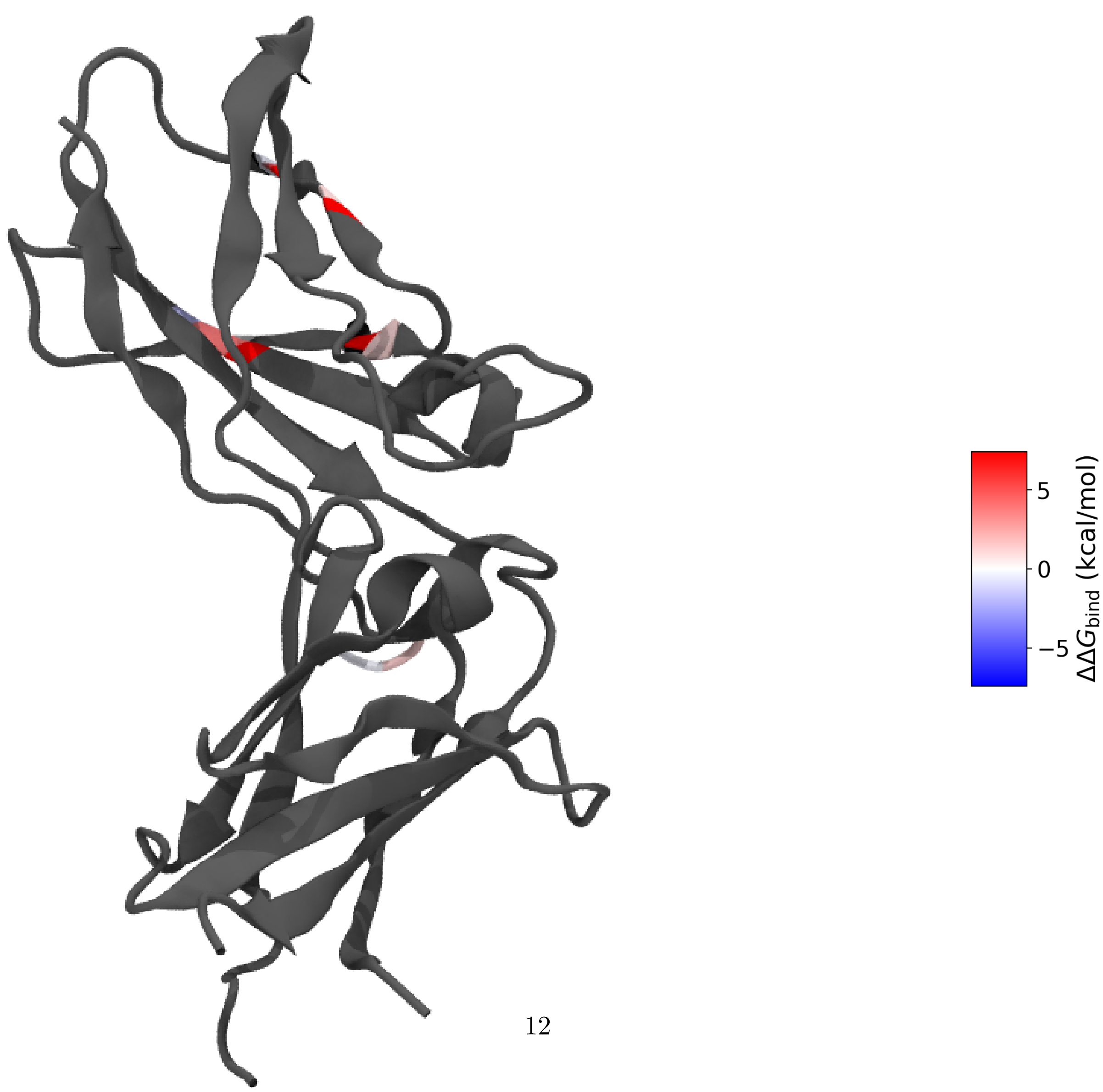
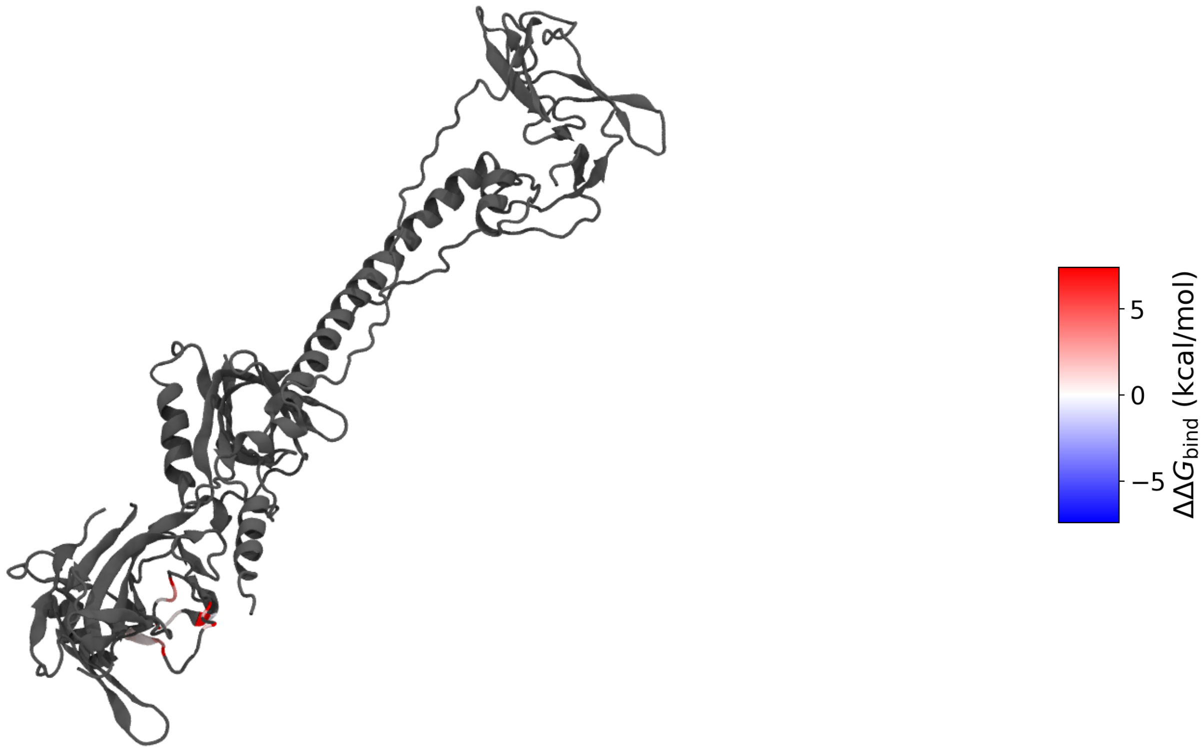


Table 7: **Tissue factor** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 1AHW chain C). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	5G9 (1AHW)
Y156	+3.81
Y157	-1.89
T167	-0.07
T170	+1.11
L176	+0.99
D178	-0.48
T197	+1.35
V198	-0.31
N199	+1.08

5.8 HCMV glycoprotein B

Glycoprotein B (gB) is the conserved class III fusion machine of human cytomegalovirus and a principal target of neutralizing antibodies. The ectodomain is a large, elongated trimer; the antibody 1G2 recognises antigenic domain 2 and its alanine scan identifies the contact hotspots (Table 8) [18]; $\Delta\Delta G_{\text{bind}}$ values curated via SKEMPI 2.0 [6]. This is our largest antigen (∼566 modelled residues) and is biologically trimeric — we simulate the apo protomer as deposited in `5C6T` (chain A), noting that the native oligomeric state may modulate the dynamics of the buried interfaces.



1G2 (5C6T)

Figure 9: HCMV glycoprotein B protomer coloured per residue by 1G2 alanine-scan $\Delta\Delta G_{\text{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned).

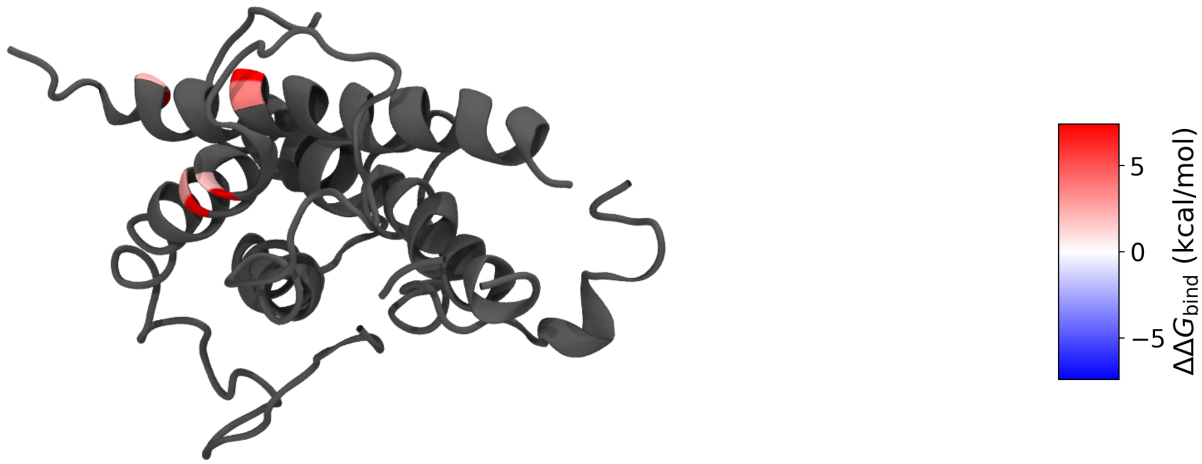
Table 8: **HCMV glycoprotein B** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; SPR, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: strip 5C6T chain A). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	1G2 (5C6T)
Y280	+3.21
N281	+0.28
T283	+0.34
N284	+0.36
N286	-0.37
F290	+2.00
E292	+1.09
F297	+2.76
F298	+0.41
I299	+0.89

5.9 Human growth hormone

Human growth hormone (hGH, somatotropin) is a 191-residue four-helix-bundle cytokine and the historical birthplace of alanine-scanning mutagenesis. Here the antibody-specific labels are the

MAb3 scan (Table 9) [5] **[verify]** (benchmark cites L. Jin's 1994 thesis, Table 3.2; the published values are in [5]); the benchmark additionally carries a secondary, provisional profile of the total $\Delta\Delta G_{\text{bind}}$ aggregated across 21 anti-hGH MAbs (not antibody-specific, held out of the primary labels). We simulate the apo hormone from `1HGU` (chain A).



MAb3 (1HGU)

Figure 10: Human growth hormone coloured per residue by MAb3 alanine-scan $\Delta\Delta G_{\text{bind}}$ (shared scale, white = 0, red = hotspot; grey = not scanned). Only the four MAb3-scanned positions are coloured.

Table 9: **human growth hormone** alanine-scan $\Delta\Delta G_{\text{bind}}$ (kcal/mol; ELISA, K_d-derived via $\Delta\Delta G_{\text{bind}} = RT\ln(K^{\text{mut}}/K^{\text{wt}})$, $T = 310\,\text{K}$; start structure: apo 1HGU). Positive $\Delta\Delta G_{\text{bind}}$ weakens binding.

Residue	MAb3 (1HGU)
R8	+2.16
R16	+3.68
D112	+1.84
D116	+2.73

6 A second label tier: binary critical-residue maps

The $\Delta\Delta G_{\text{bind}}$ antigens above are the gold standard — a graded, energetic importance per residue — but experimental alanine scans with full $\Delta\Delta G_{\text{bind}}$ tables are scarce. A much larger body of data exists at a coarser but still *energetic* resolution: **saturating shotgun alanine scans** read out by flow cytometry (Integral Molecular's platform; Davidson & Doranz [19]), which classify each scanned residue as *critical* or *non-critical* for a given antibody. A residue is critical when its mutation measurably abolishes binding, so even this binary call is grounded in binding energy, not mere proximity — it is the same quantity as $\Delta\Delta G_{\text{bind}}$, thresholded. We therefore treat these as a second label tier: **Tier 1**, graded $\Delta\Delta G_{\text{bind}}$ (§5); **Tier 2**, binary critical/non-critical.

Source and the proximity filter. We mine these from IEDB (`query-api.iedb.org`) [1], keeping only conformational (discontinuous) epitopes defined by *binding* assays and explicitly discarding every structure/contact-derived record — the proximity labels this project argues against (recall that $>$ 99% of structure-linked IEDB epitopes are crystal/cryo-EM contact, §1). Within the binding-derived set we take the flow-cytometry (shotgun-mutagenesis) subset, whose curated residue list *is* the set of critical residues. This yields 3,213 antibody–antigen epitope maps over 186 antigens. Table 10 lists the simulatable, not-yet-included candidates; note the depth of antibody sampling per antigen (e.g. dengue envelope: 781 mapped antibodies across 30 studies), which lets a Tier-2 antigen carry a *consensus* epitope (residues critical across many antibodies) as well as antibody-specific maps.

Table 10: **Tier-2 (binary) label candidates** from IEDB shotgun-mutagenesis alanine scans (flow-cytometry critical-residue mapping; Integral Molecular / Doranz). Labels are per-residue *critical vs non-critical*, energetically grounded (a critical residue is one whose mutation measurably weakens antibody binding) but coarser than $\Delta\Delta G_{\text{bind}}$. "Antibodies" = mapped antibody footprints (one antibody → one conformational epitope); "Studies" = distinct source papers; "Crit. res." = union of critical residues across all antibodies. Structures provisional ([v] = numbering to be verified before use).

Antigen	Organism	Antibodies	Studies	Crit. res.	Structural handle
Dengue E (envelope)	Dengue virus	781	30	632	sE ectodomain, e.g. 1OKE [v]
Chikungunya E1/E2	Alphavirus chikungunya	128	6	98	ectodomain, e.g. 3N42 [v]
RVFV Gn	Rift Valley fever virus	63	8	210	Gn ectodomain, e.g. 6F9F [v]
Influenza HA	Influenza A virus	81	22	296	HA ectodomain (trimeric) [v]
GRO-α / CXCL1	Homo sapiens (human)	30	1	23	chemokine, e.g. 1MGS [v]
CD4	Homo sapiens (human)	1	1	18	D1–D2, e.g. 1CDH [v]
CD40L	Homo sapiens (human)	1	1	13	TNF domain, e.g. 1ALY [v]
IL-2Rα	Homo sapiens (human)	2	1	13	ectodomain, e.g. 1Z92 [v]
IL-7Rα	Homo sapiens (human)	2	1	32	ectodomain, e.g. 3DI2 [v]
HLA class I	Homo sapiens (human)	15	5	38	peptide-MHC, many [v]

Numbering verification. The shotgun residues are reported in UniProt numbering and carry their wild-type identity, but aggregating by UniProt accession mixes proteins, numbering conventions, and (for viruses) serotypes/subtypes. We therefore map every residue onto a single reference structure and *discard any residue whose wild-type identity does not match* — exactly the wild-type check applied to the $\Delta\Delta G_{\text{bind}}$ antigens. This is strict: of dengue's raw critical positions only those matching DENV2 envelope survive, and likewise only H3 residues survive for influenza. What remains is clean and biologically coherent — the top dengue residue (fusion loop, E101) is called critical by 102 of 632 antibodies.

Representation. We render each staged antigen two ways (Figs. 11, 12): **binary** (red = critical for at least one antibody, grey = not) and **consensus frequency** (white→red by the number of antibodies for which the residue is critical). The frequency view is itself an energetic-importance signal — it ranks residues by immunodominance across hundreds of antibodies — and reduces to the binary view under any threshold, so we keep both and decide later. Tier-2 antigens enter the benchmark as a separate (binary/graded) label stream, never merged into the $\Delta\Delta G_{\text{bind}}$ tables; in the leakage-controlled evaluation they are held out by antigen.

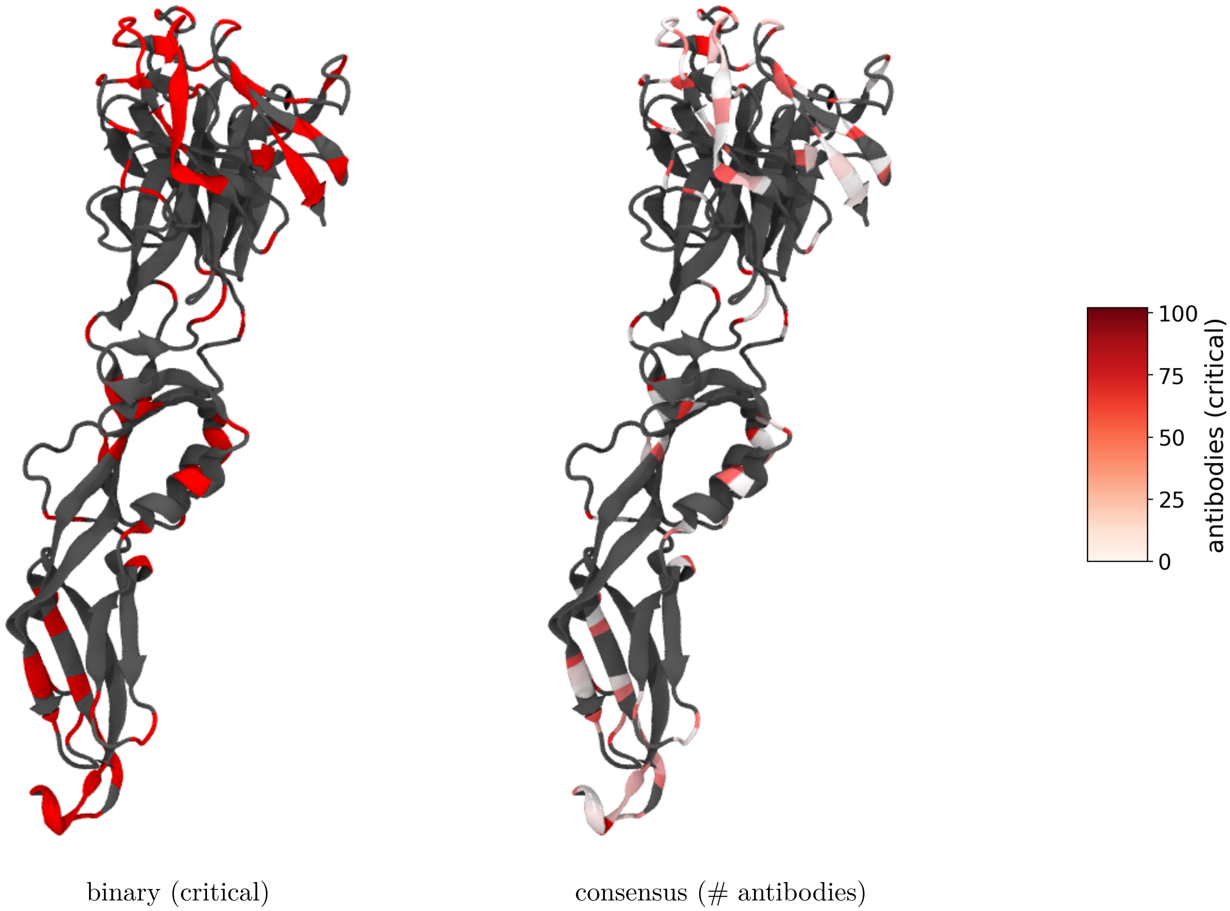


Figure 11: **Dengue envelope (DENV2, `1OKE` sE):** 96 wild-type-verified critical residues from 632 antibodies. Binary (left) and antibody-frequency consensus (right). The deepest red is the conserved fusion loop (E101, critical for 102 antibodies); a second focus is the EDIII lateral ridge — both textbook dengue neutralization sites.

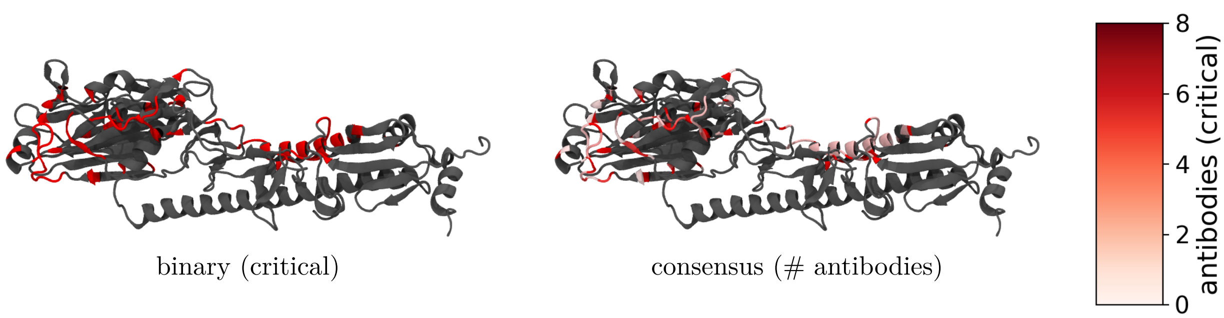


Figure 12: **Influenza hemagglutinin (H3 X-31, `1HGG` protomer):** 84 wild-type-verified critical residues (HA1 head + HA2 stem) from 79 antibodies, binary and antibody-frequency consensus. Critical residues concentrate on the HA1 globular head (the classic antigenic sites) with a sparser stem contribution. Native HA is an obligate trimer; one protomer is shown.

7 Static baselines: exposure and mobility

Before adding any dynamics, we ask how far purely *static*, single-crystal descriptors already get — both to set the bar the dynamics features must clear and to make the proximity critique of §1 quantitative on our own labels. We test the two classic static proxies for "epitope-ness": surface **exposure** (this section) and **mobility** (crystallographic B-factors, below). The exposure descriptor is per-residue **relative solvent-accessible surface area** (rel. SASA): Shrake–Rupley SASA on the apo structure, normalised by the residue-type maximum [20]. This is a single-frame calculation on the apo input structures — the pre-trajectory baseline; once the production runs land it is replaced by trajectory-averaged SASA at the same residue keys (and joined by RMSF, non-bonded energetics, and hydration).

Setup. We pool every residue of the eight $\Delta\Delta G_{\text{bind}}$ antigens (1,790 residues). A residue is *important* if its alanine-scan $\Delta\Delta G_{\text{bind}} \geq 1$ kcal/mol for any antibody (aggregated as the max over antibodies). An exposure-only predictor labels a residue an epitope if rel. SASA ≥ 0.25 (the standard "exposed" cutoff) — the static analogue of a proximity label. We then read off precision and recall against the experimental importance labels.

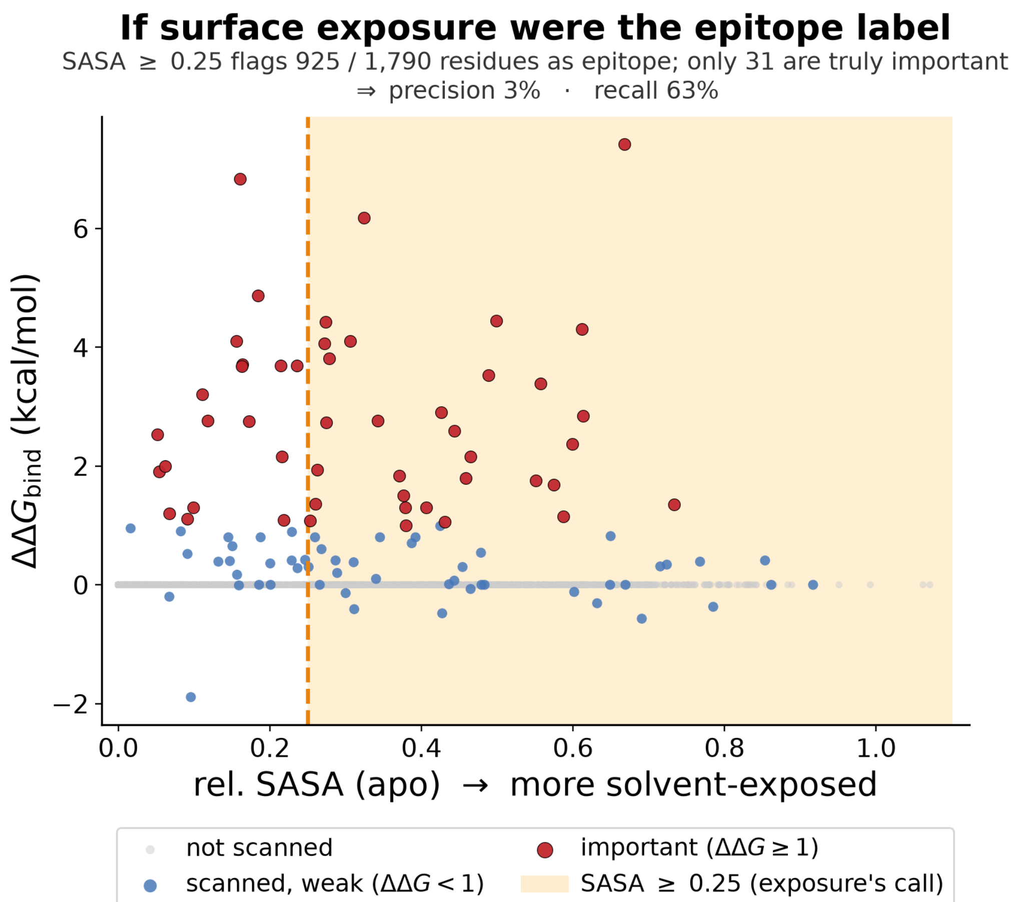


Figure 13: **Surface exposure as an epitope label.** Every residue of the eight $\Delta\Delta G_{\mathrm{bind}}$ antigens, apo rel. SASA (x) vs. alanine-scan $\Delta\Delta G_{\mathrm{bind}}$ (y); red = scanned and important ($\Delta\Delta G_{\mathrm{bind}} \geq 1$), blue = scanned but weak ($\Delta\Delta G_{\mathrm{bind}} < 1$, incl. binding-improving), grey = not alanine-scanned (no data, plotted at 0); the shaded band is what an exposure cutoff (rel. SASA ≥ 0.25) would call epitope. Exposure flags 925 of 1,790 residues as epitope but only 31 are truly important (precision $\approx 3\%$), while still missing the buried-at-interface hotspots to the left of the line (recall $\approx 63\%$). Single-frame apo SASA; `analysis/compute_sasa.py` + `plot_feature_vs_label.py`.

Result. Exposure behaves exactly as the thesis predicts. It has *moderate recall* (it knows roughly where the surface is: 63% of importance hotspots are exposed) but *near-zero precision* ($\approx 3\%$): rel. SASA ≥ 0.25 sweeps in almost half the protein, overwhelmingly residues that contribute no binding energy. Per antigen the precision is 0–14% (highest for the clean single-domain antigens `1AKI`, `1BJ1`, `1JRH`). And *among the scanned interface residues*, exposure does not even rank graded importance: the pooled Spearman correlation between rel. SASA and $\Delta\Delta G_{\mathrm{bind}}$ is -0.16 — once a residue is on the surface, how exposed it is says nothing about how much binding energy it carries. That gap — high recall, no precision, no within-interface ranking — is precisely what the dynamic and energetic features are meant to fill, and every feature we add is scored against this baseline.

Caveats. (i) Single-frame apo SASA; trajectory-averaging will move values and is the real control. (ii) `5C6T` (gB) is anomalous here — it is scored on the unrepaired structure (§9) *and* as an isolated protomer of a native trimer, so residues buried in the trimer interface read as exposed; `2JEL` (HPr) carries only one hotspot. (iii) Residues never scanned are treated as non-epitope, which alanine scans largely justify (they target the known epitope) but which inflates the negative set.

Static mobility (crystallographic B-factors) — the dynamics teaser. A second, long-standing idea links antigenicity to *flexibility*: mobile surface regions are more antigenic [2]. The static, experimental proxy for residue mobility is the crystallographic B-factor, and it is the *static analogue of the RMSF* we will extract from the apo trajectories — so a B-factor signal would foreshadow the dynamics payoff. We test it the same way: per-residue Cα B-factor, z-normalised per structure (raw B-factors are incomparable across crystals — resolution, refinement, overall scaling), against $\Delta\Delta G_{\text{bind}}$ (Fig. 14). The result runs *opposite* to the naive expectation. The energetic hotspots are among the most *rigid* residues, not the most mobile: pooled Spearman $\rho = -0.29$, negative in six of eight antigens, with whole-protein AUC < 0.5 (anti-discriminative). Residues that pay the most binding free energy are pre-organised, low-B anchors, not floppy high-B loops — and the classic flexibility–antigenicity correlation concerns where antibodies *bind* (surface loops), not which residue carries the binding *energy*, exactly the proximity-vs-energetics distinction of §1.

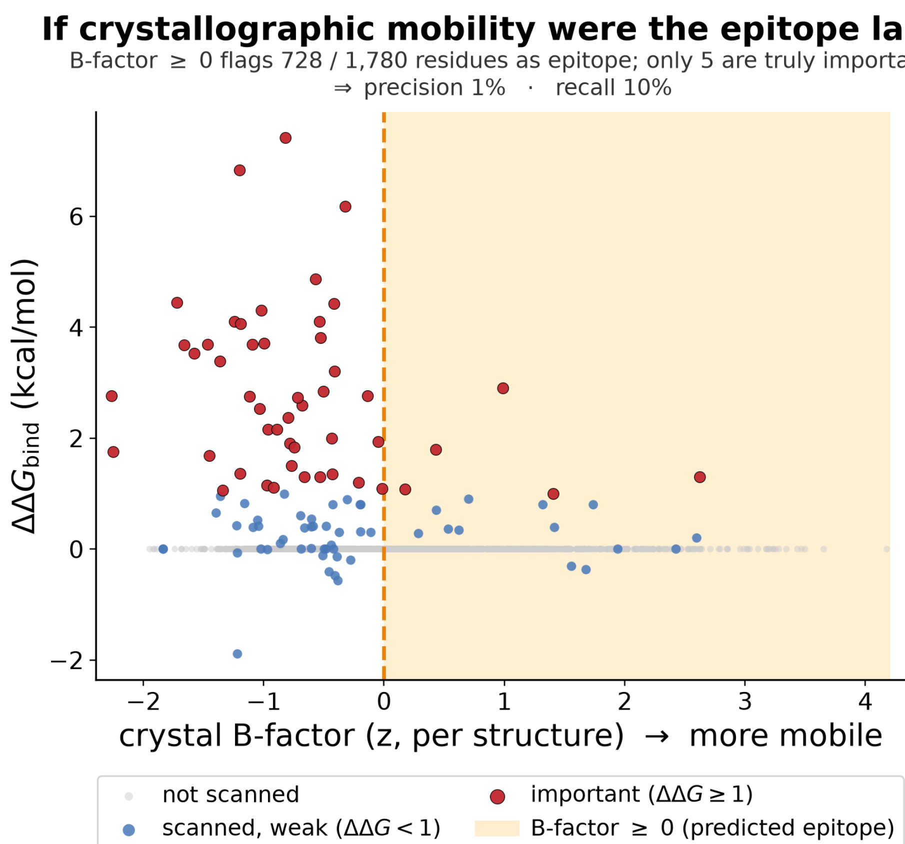


Figure 14: **Crystallographic mobility as an epitope label.** Per-residue Cα B-factor (z-scored per structure; right = more mobile) vs. alanine-scan $\Delta\Delta G_{\text{bind}}$, colours as in Fig. 13. The important (red) residues cluster at *low* B-factor (left, rigid), not in the high-mobility band — crystal mobility inversely tracks energetic importance (pooled Spearman -0.29). `analysis/compute_bfactor.py` + `plot_feature_vs_label.py`.

Where this leaves us. Neither static proxy recovers the energetic hotspots from a single crystal: exposure has no precision ($\sim 3\%$), and mobility points the wrong way ($\rho = -0.29$). That is precisely the opening for apo-MD — RMSF is a cleaner, crystal-contact-free mobility measure, and the trajectory additionally exposes the correlated-motion, mode-participation (FRESEAN), and hydration (3D-2PT) features that no single structure carries. The question the rest of this work asks is whether dynamics *done properly* recovers signal where these static proxies fail. (Caveat: B-factors are contaminated by crystal contacts, resolution, and refinement protocol; the per-structure z-score removes only the overall scale.)

8 Methods — apo-HEL run #1

Frozen, signed-off stack in `../docs/METHODS.md` (2026-06-18). Purpose of run #1: *validate the prep→equil→production pipeline on Gemini and extract a first RMSF / flexibility feature* (the Westhof re-test). House default stack signed off as the starting point.

Stack. AMBER99SB-ILDN / TIP3P; cubic box, 1.2 nm clearance; neutralize + 0.15 M NaCl; leapfrog, 2 fs, LINCS on h-bonds; PME ($r_{\text{coul}} = r_{\text{vdw}} = 1.0$ nm, potential-shift, DispCorr); V-rescale at 300 K (Protein / Non-Protein); EM → 100 ps NVT → 100 ps NPT (both position-restrained) → **100 ns** production (single trajectory, for RMSF statistics). Random seeds (`gen_seed` = `ld_seed` = −1): bitwise reproducibility is not the target — the workflow reproduces from git-tracked inputs and saved derived outputs, and random seeds give genuinely independent replicas. Frozen per-run `.mdp` files: `md/1AKI/apo/configs/` (run dirs are organized `md/<PDB>/<state>/`).

Deliberate deviations from the house scripts. (i) **C-rescale** barostat for NPT equilibration (Parrinello-Rahman only in production), since PR can ring when started far from equilibrium; (ii) **cubic** box (the old house scripts disagreed cubic/dodecahedron) — chosen for interpretability and uniform across all systems, at a modest extra-solvent cost vs. dodecahedron.

HEL-specific checkpoints (not glossed over). pdb2gmx must form HEL's **4 disulfides** (6–127, 30–115, 64–80, 76–94) — verify the log reports 4 SS bonds. His15 protonation: record the assigned state. Crystal waters (78 HOH) removed; fresh TIP3P added. Read every grompp warning rather than letting `-maxwarn` hide it.

Caveats / flags. TIP3P over-mobilizes water and 2 fs+LINCS suppress H dynamics — fine for relative RMSF and pipeline validation, *not* for the dynamics-grade features. 100 ns is a single trajectory for statistics; replicas are deferred (random seeds make adding independent ones trivial).

Deferred: the distinctive run. FRESEAN mode participation and 3D-2PT hydration thermodynamics require a *separate* protocol — high-frequency velocity output, constraint removal / sub-fs dt, TIP4P/2005 — with settings pulled from the FRESEAN and energy-transfer manuscripts. This is the run that powers the mode-based and hydration feature classes.

9 Structure completeness and the repair strategy

Several crystal structures are incomplete — disordered termini, internal loops, or whole segments that the experiment never resolved (PDB `REMARK 465`). Apo-MD demands a chemically complete chain, so each gap forces a choice. We make that choice explicitly per structure rather than feeding everything through one tool, because **the right repair scales with gap length**, and the wrong tool on a long gap produces a confident-looking but meaningless model.

What a "fixer" can and cannot supply. The instinct to "just grab the FASTA for the missing residues" conflates two different problems. The deposited sequence (`SEQRES`, the same residues a fixer recovers automatically by diffing `SEQRES` against the resolved `ATOM` records) supplies the *identity* of the missing residues — but identity is the easy half. The hard half is *geometry*: where the missing backbone atoms sit in three dimensions, about which the sequence says nothing. `pdbfixer` (OpenMM) builds missing heavy atoms onto resolved residues from templates — valid and routine — and threads missing residues in with default internal coordinates followed by a short minimization of only the new atoms. That minimization removes clashes; it performs *no* conformational sampling. The result is therefore reliable for a short surface gap (apo equilibration relaxes a two- or three-residue loop anyway) and arbitrary for anything longer, where real loop modelling (Modeller, Rosetta) or a full predicted model (AlphaFold) is required. Disordered *termini* are not modelled at all: they bear no labels and would only add unrestrained flailing, so we cap them.

Method used: AlphaFold-loop grafting with restrained closure. Rather than let a fixer build loops blind, we graft the missing internal residues from an AlphaFold model and use the crystal for everything resolved. For each antigen we (i) sequence-align the crystal chain to its AlphaFold model to fix the numbering offset, (ii) measure the Cα-RMSD between model and crystal over all matched residues — the "is the prediction trustworthy here" test — and (iii) for each internal gap, locally superpose the model on the flanking resolved residues and splice in its loop at crystal numbering. `pdbfixer` then completes any partial side chains, and a short OpenMM minimization with all crystal atoms position-restrained (only the loop free, plus an explicit harmonic across the spliced junction) closes the peptide bonds. Disordered termini are capped, not modelled. Scripts: `analysis/repair_structure.py`, `finalize_structure.py`, `close_loops.py`; host-side env `env/repair-environment.yml` (Python 3.11, OpenMM 8.5, `pdbfixer`, run locally, not on Gemini).

Antigen (PDB, chain)	Unresolved segments (`REMARK 465`)	Repair outcome
hGH (`1HGU` A)	F1; **K38–E39** (internal); F191	AF-graft (AFDB `P01241`); **done**
Tissue factor (`1AHW` C)	S1–T3; **V83–G90** (8-res); Q212–E219	AF-graft (AFDB `P13726`); **done**
HCMV gB (`5C6T` A)	63–87 (25); 115–120; 219–220; 237–239; **437–468 (32)**	not in AFDB → AF3 prediction (pending)

For tissue factor the AlphaFold model agrees closely with the crystal (global Cα-RMSD 1.12 Å over 200 residues; the V83–G90 loop fits its flanks at 1.29 Å), so the 8-residue graft is well supported. For hGH only the 2-residue K38–E39 turn is grafted (local fit 1.81 Å); the larger global RMSD (3.36 Å, hinge flexibility of the four-helix bundle) is immaterial for so short a splice. Both repaired chains validate as continuous, with the expected disulfides (hGH 53–165, 182–189; TF 49–57, 186–209) and no steric clashes. HCMV gB is a viral protein *absent from AlphaFold DB*, and its largest gap (32 residues) is true *de novo* prediction, not loop repair; it requires a fresh model. We generate one with **AlphaFold3** on Gemini (the lab's `alphafold_3.0.1` Singularity container and installed weights/databases; `md/5C6T/apo/af3.sbatch`, mirroring the episcaf pipeline), then graft its loops the same way. The AF3 model's agreement with the crystal over the resolved residues is the same trust check applied above; if it disagrees we simulate the AF3 ectodomain directly.

Consequences for the run queue. Eight of the nine non-pristine/pristine antigens are now ready to prep on Gemini (`git pull` then `sbatch`): pristine chains HEL (`1AKI`), HPr (`2JEL`), VEGF (`1BJ1`), IFN-γR (`1JRH`), Dengue E (`1OKE`), HA (`1HGG`); plus the repaired hGH (`1HGU`) and tissue factor (`1AHW`). Outstanding: HCMV gB (`5C6T`, awaiting an AF3 model) and the previously deferred MT-SP1 (§5, two internal residues, same AF-graft recipe) and Bont/A1.

10 Run log

Exact-command reproducibility ledger. Append a row per stage; pin the GROMACS version actually used on Gemini, seeds, and Slurm job id.

Date	Stage	Command	GROMACS ver.	Job / output
2026-06-18	structure	`curl .../1AKI.pdb`	—	`structures/1AKI.pdb`
pending	prep	`sbatch md/1AKI/apo/prep.sh`	2023.2-dev	out/npt.gro/cpt
pending	production	`sbatch md/1AKI/apo/md.sh`	2023.2-dev	out/md.xtc

Status (2026-06-18). Project scaffolded; benchmark audited; first system (HEL/1AKI) selected and label mapping verified 1:1; apo-HEL run #1 staged and signed off. Gemini environment confirmed: module `Gromacs` (capital G), GROMACS `2023.2-dev` (avx2_256), partition `gpu-a100` (4-day limit, A100 fastest); repo cloned to `/scratch/bneff/bcell_epitope`. Production walltime set to 48 h ceiling. **Next:** `git pull` on Gemini, then `sbatch prep.sh` → `sbatch md.sh`; record Slurm job ids and ns/day. Production length 100 ns (single trajectory). Open: replicas (deferred — random seeds make them trivial); scheduling the FRESEAN/3D-2PT run.

Run-script fixes (bringing up run #1 on Gemini). Three issues surfaced and were fixed while getting prep to run; recorded for reproducibility:

- **Module name.** Gemini's module is `Gromacs` (capital G); lowercase `gromacs` errors on a missing `shared` dependency. Scripts updated.
- **Config paths under Slurm.** Slurm copies the batch script to a private spool dir, so `$(dirname $0)/configs` resolved into the spool dir and `ions.mdp` was not found. Scripts now anchor paths to `$SLURM_SUBMIT_DIR`; **always submit from the repo root**.
- **GPU bonded offload on EM.** `-bonded gpu` requires a dynamical integrator; EM uses `steep`, so the EM `mdrun` now uses `-nb gpu` only. NVT/NPT/ production use `md` and keep `-nb/-pme/-bonded gpu` (audited).

Also removed the obsolete `ns_type` entry from all mdp files (ignored under the Verlet scheme). **Caveat:** Gemini scratch is not durable — move logs/outputs off scratch after each run (an earlier prep's logs were lost to a purge). Sanity check per run: `pdb2gmx.log` must report **4 disulfides** for HEL.

Expansion to multiple antigens (2026-06-21). Switched the box from dodecahedron to **cubic** (interpretability; uniform across systems) and re-ran HEL for consistency. Staged three further apo-antigen systems from the benchmark, hybrid-sourced (strip antigen chain(s) from the complex; numbering kept aligned to the labels). Provenance:

- **HPr** — `2JEL` chain P (res 1–85, no Cys); `md/2JEL/apo`.
- **VEGF** — `1BJ1` chains V+W (res 14–107, disulfide-linked homodimer; `pdb2gmx -merge all` for the inter-chain SS); `md/1BJ1/apo`.
- **Bont/A1 Hc** — `2NYY` chain A res 872–1295 (C-terminal receptor-binding domain only; catalytic-domain Zn/Ca excluded; res 872 is a domain cut → charged N-terminus); `md/2NYY/apo`.

All four (HEL+these) submitted overnight as prep→production chains (`sbatch --dependency=afterok`). **Deferred:** MT-SP1 (2 missing internal residues), Bont/A2 (homology models), hGH (provisional labels). Per-system disulfide expectations recorded in each `prep.sh`.

Structure repair (2026-06-29). Audited `REMARK 465` for the incomplete staged structures and repaired two of three by AlphaFold-loop grafting with restrained closure (method + RMSDs in §9; scripts `analysis/repair_structure.py`, `finalize_structure.py`, `close_loops.py`; host env `env/repair-environment.yml`). **hGH** (`1HGU`, K38–E39 grafted from AFDB `P01241`) and **tissue factor** (`1AHW`, loop V83–G90 from AFDB `P13726`; AF–crystal global Cα-RMSD 1.12 Å) are now frozen as `structures/{hGH_1HGU,TissueFactor_1AHW}_fixed.pdb` — continuous, correct disulfides, clash-free — so both are ready to prep. **HCMV gB** (`5C6T`) is *absent from AlphaFold DB* (viral) and its 32-residue gap is de-novo prediction; deferred pending an AF3 (AlphaFold Server) model. Net: eight antigens ready to `sbatch` on Gemini.

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